Probabilistic dengue and chikungunya forecasting in Brazil: metric-dependent model rankings, regime-shift fragility, and conformal ensembles

Md Kamruzzaman Kamrul[1], Eashraque Jahan Easha[2], and Abdullah Al Helal[3]

[1]Purdue University, West Lafayette, IN, USA
[2]University of Denver, Denver, CO, USA
[3]Oklahoma State University, Stillwater, OK, USA

November 9, 2026

Abstract

Dengue is expanding across Brazil; 2024 produced a record epidemic roughly four times any prior year. Accurate, calibrated probabilistic forecasts support preparedness, but which of statistical, machine-learning, deep-learning, and mechanistic approaches forecasts best, especially under an outbreak year's distribution shift, is unsettled. We report a reproducible, leakage-controlled comparison of five probabilistic forecasting model families, 25 models and variants in total, for the 3rd Infodengue–Mosqlimate Dengue Challenge (IMDC 2026), which requires weekly, state-level, full-interval forecasts for the 26 mandatory Brazilian states. All models are evaluated through a single backtesting harness over four official retrospective seasons, one of which contains the 2024 outbreak, and scored by the weighted interval score (WIS). We report four results with implications beyond this challenge. First, no model is best across seasons: global recurrent networks are the most accurate single models in normal seasons, yet the one we deployed is the worst of all on the 2024 outlier, while an ensemble never incurs any member's worst-season failure. Second, the best model depends on the evaluation metric: magnitude-weighted mean WIS and scale-free normalized WIS reward accuracy on different subsets of a highly skewed set of states. Third, a conformal recalibration, a post-hoc widening of the ensemble's intervals tuned to fix their overconfidence, improves accuracy and calibration at negligible cost (mean WIS, a magnitude-weighted score, 1234 to 1162; normalized WIS, its scale-free counterpart, 0.595 to 0.561; out-of-sample reduction of 6.2 percent), acting as asymmetric insurance: little cost in an ordinary season, but it prevents the catastrophic under-prediction an extreme one would otherwise cause. The right recalibration is itself metric-dependent: scored state by state against the challenge organizers' reference model, the single conformal factor over-widens low-burden states, and a size-aware variant that scales each state's intervals by its historical burden keeps per-state skill (geometric-mean WIS ratio to the reference 0.916, against 0.989 for conformal) while recovering 85 percent of the conformal gain in mean WIS. Fourth, added model complexity did not pay: observed-climate covariates and hierarchical spatial pooling both failed to improve forecasts, and an internal-holdout hyperparameter gain reversed on the true backtest. The best model is disease-specific: for chikungunya, gradient boosting alone

outperforms the ensemble. All code, the backtesting harness, per-state scored predictions, and validated submissions for four official tracks are released, with an automated test suite covering leakage, scoring, and every model.

Author summary

Brazil carries one of the world's largest dengue burdens, and 2024 brought an epidemic without precedent in the national record. Public-health agencies increasingly rely on forecasts that give a full range of plausible outcomes with calibrated uncertainty, not just one predicted number. Which method to trust is an open question, especially in an exceptional season, when accurate forecasts matter most. We compared five model families, from simple seasonal baselines to gradient boosting, deep learning, and a mechanistic epidemic model, using identical retrospective tests across four dengue seasons in all 26 mandatory Brazilian states. No method is best everywhere: deep learning was the most accurate approach in ordinary seasons but failed badly during the 2024 outbreak, while a simple ensemble that averages the models was the most dependable. The apparent winner also changes depending on how performance is scored, a caution for forecasting leaderboards. Finally, a light statistical recalibration that stretched the ensemble's prediction intervals just enough to fix their overconfidence improved both accuracy and reliability. Adding more data or model complexity did not help. At the challenge's own validation-round results presentation, our submission was independently named one of six best-overall-performing models (of 27 submitted) for the mandatory state-level dengue target, and one of three models with outstanding performance on the optional city-level target [36]. Our full pipeline is openly available.

1. Introduction

Dengue is among the fastest-spreading mosquito-borne diseases, with an estimated 390 million infections each year and roughly half of the world's population at risk, a range that continues to expand with climate change and urbanization [27, 29, 30]. Brazil bears one of the largest national burdens [18]. The 2024 season broke the national record by a wide margin, with a peak of 451,578 reported cases in a single epidemiological week and an annual total near 6.67 million, roughly four times the next-largest year on record. Transmission is shaped by climate, and the 2023–24 El Niño is one of several drivers implicated in the surge. Against this background, arboviral forecasting has moved from point prediction toward full probabilistic forecasts, motivated by the recognition that preparedness decisions depend on the entire predictive distribution rather than a single expected value [19, 23, 32].

The Infodengue–Mosqlimate Dengue Challenge (IMDC) operationalizes this shift. It invites teams to submit weekly, state-level probabilistic forecasts with prescribed quantiles, scored by the weighted interval score, following the precedent of the inaugural 2024 edition [1]. The challenge exposes a question that the wider forecasting literature has not resolved for this setting: given statistical, machine-learning (ML), deep-learning (DL), and mechanistic approaches, which forecast best, and how robustly, when a season can depart by an order of magnitude from anything in the training record?

Prior work on dengue forecasting is extensive but leaves that question open in three specific ways. First, most published comparisons rank models on point accuracy. A systematic review and

network meta-analysis of 59 dengue-forecasting studies from 2014 to 2024 ranked methods on root-mean-square error and found simple statistical models competitive with machine learning at lower data and implementation cost [6]. Point accuracy does not measure whether a prediction interval is calibrated, which is what a preparedness decision needs. Second, the probabilistic comparisons that do exist mostly target short horizons. The largest, an evaluation across more than 180 locations worldwide, forecasts one to three months ahead and finds ensembles consistently among the top performers [3]; a Brazilian statistical model forecasts probabilistic epidemic bands, the activity range a coming season is expected to fall within, rather than a weekly trajectory [4]. A full-season target, here 16 to 67 weeks ahead because a 15-week reporting gap separates the training cutoff from the first target week, is a different problem: no recent case data is available near the target, so the models that win at one month need not win here. Third, the inaugural sprint reported that no single model excelled across locations and years, and observed that its three scoring rules did not agree on a winner [1]. The question we take up is a different one: not whether separate scoring rules disagree, which is expected of rules measuring different properties, but whether two aggregations of the *same* proper scoring rule, one weighted by case magnitude and one scale-free, select different models. That disagreement cannot be attributed to the scores measuring different things, and it determines what a leaderboard reports.

We address this with a controlled comparison built for reproducibility and for honest evaluation under distribution shift. Every model is fit and scored through one backtesting harness that enforces leakage safety and applies the same scoring code to all forecasts, over the four official retrospective seasons (2022–23 through 2025–26) for the 26 mandatory states. The design trades breadth for depth relative to multi-country evaluations [3]: one country, two diseases, a full-season horizon, and every model sharing one leakage-audited pipeline whose tests, provenance manifest, and per-state scored predictions are public. That depth is what makes the negative results and the metric-disagreement finding below checkable rather than anecdotal. We make the following contributions.

1. We show that model rankings are season-dependent and, in the outbreak year, invert: the model that is best on average can be the least reliable exactly when reliability matters most (Section 2.2).

2. We show that the identity of the best model depends on whether skill is weighted by case magnitude or by forecasting unit, a consequence of real cross-state heterogeneity rather than an artifact, and we argue for reporting both (Section 2.5).

3. We introduce a conformal recalibration of the ensemble intervals that improves accuracy and calibration at negligible cost and functions as asymmetric insurance against catastrophic under-prediction (Section 2.3), and show that a size-aware variant is preferable when each state's forecast counts equally (Section 2.14).

4. We document informative negative results: added covariates, hierarchical pooling, hyperparameter tuning, and an alternative ensemble-combination rule each failed to improve forecasts where it mattered most, which clarifies where the predictable signal in this problem lies (Section 2.9).

5. We show that the best model is disease-specific, using chikungunya as a contrasting target (Section 2.10), and we release the complete, tested pipeline.

2. Results

2.1. Study design and data

We forecast weekly dengue case counts per state across the 2022–23 through 2025–26 seasons for the 26 mandatory Brazilian states (Espírito Santo is excluded per the challenge specification). These four seasons are referred to throughout as folds 1 through 4 (fold 1: 2022–23; fold 2: 2023–24, containing the 2024 outbreak; fold 3: 2024–25; fold 4: 2025–26, partially resolved). Each fold is defined by an official training cutoff, the last date a model fitted for that fold is allowed to see, and a target window, the season it must forecast; a confirmed 15-week gap separates each training cutoff from the start of its evaluation window, reflecting the reporting delay a real forecaster faces (Fig 1). Forecasts are submitted as nine quantiles per state and week, from which the four central prediction intervals (50, 80, 90, and 95 percent) are formed. All covariate access, feature construction, and model fitting are filtered to data at or before each training cutoff, and this discipline is enforced by automated tests rather than by convention (Section 4). The four seasons differ by an order of magnitude in difficulty. The 2023–24 season (fold 2), which contains the 2024 outbreak, is the hardest for every model in Table 1, and the 2025–26 season (fold 4) is only partially resolved in the available data and is reported separately.

2.2. No single model dominates across seasons

Table 1 reports mean WIS by model and season, and Fig 2 shows the same on a log scale. The ranking of models reorders across seasons. The deep network (a global GRU with a negative-binomial head, an output layer suited to over-dispersed count data such as case counts, rather than the Poisson head more common in count models) is the strongest single model in the two ordinary seasons, leading fold 1 outright at WIS 348 and placing second on fold 3 at 324, behind only its own pinball-trained variant at 321 (Section 2.8); the three ensembles beat it on fold 1, which is the point of combining. It is nonetheless the worst of all models on the 2024 outlier, fold 2 (WIS 4131), worse even than a naive baseline, because the surge lies outside its training distribution and its intervals, derived from its fitted distribution instead of built empirically, are too narrow to cover it. The gradient-boosted model (XGBoost, chosen over a second gradient-boosting implementation, LightGBM, that we also tested and that scored significantly worse overall, Section 4) is the strongest of the three deployed families on the outlier season (WIS 3238), and unlike LightGBM, which showed a sharper trade-off (strongest of the two on the outlier at WIS 3332 yet the weakest of the serious models on the ordinary season fold 3, WIS 665), XGBoost remains reasonably competitive instead of collapsing in the ordinary seasons (WIS 376 and 484 in folds 1 and 3). Two models outside the deployed ensemble handle the outlier season better still: the surge template at WIS 3118 and the semi-mechanistic trajectory model at 3371, both for the same reason, which is that each builds its intervals by resampling whole historical seasons and so widens them precisely when the target departs from the recent past (Section 2.8). This season-dependence is the central obstacle the challenge poses: a model selected on average performance carries a hidden risk of catastrophic failure in the season that matters most.

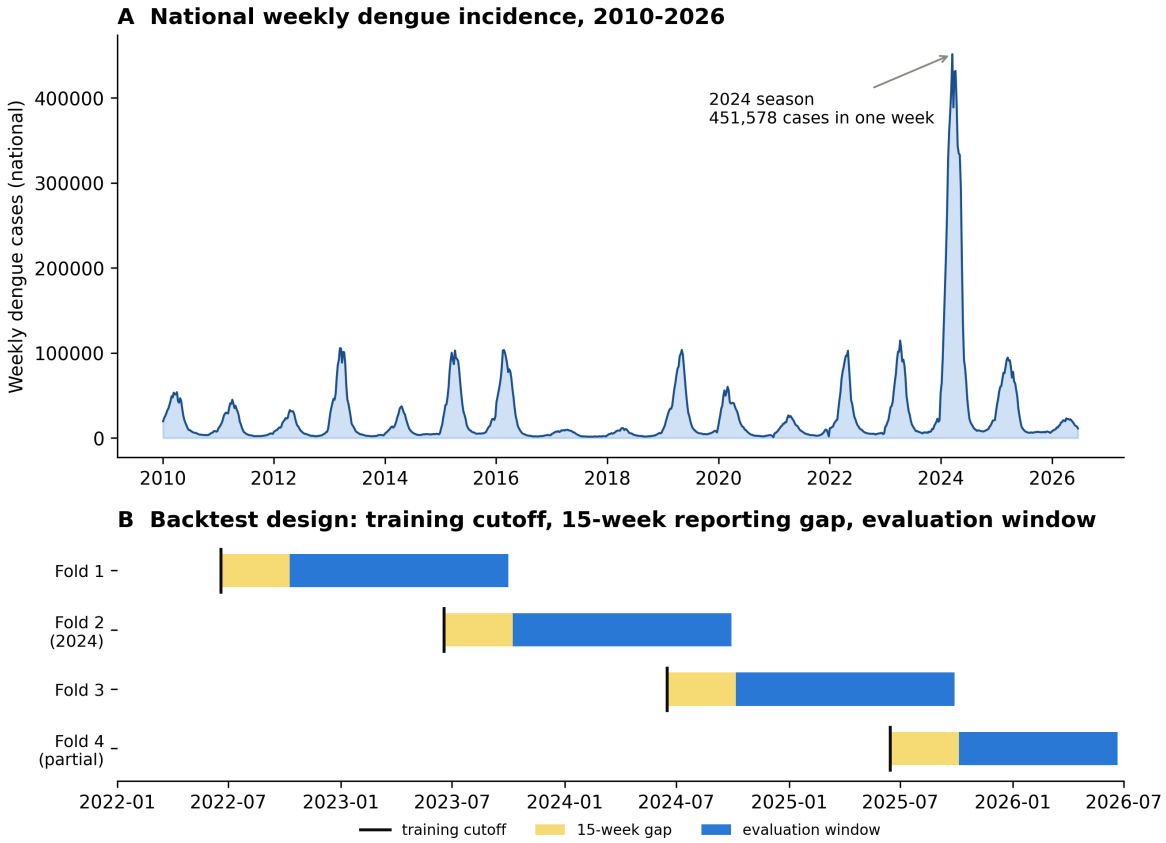


Figure 1: Data and study design. (A) Weekly reported dengue cases aggregated to the national level, 2010–2026; the 2024 season stands roughly four times above any prior year. (B) The four official retrospective seasons form a rolling backtest. Each has a training cutoff (all data up to the cutoff is used for fitting), a 15-week reporting gap, and an evaluation window on which forecasts are scored. Fold 2 covers the 2024 outbreak and fold 4 is partially resolved.

Season-to-season stability is a distinct axis from mean skill, and the two disagree for the models identified above. For each fold we compute each model's relative WIS against the seasonal-naive baseline (a fully-crossed pairwise comparison over state and horizon, Cramer et al. 2022), take its log so that a model twice as good and a model twice as bad are symmetric, and summarize the four folds by their mean (overall skill against the baseline, negative is better) and standard deviation (season-to-season consistency, lower is more stable). Fig 3 plots the two against each other. A log relative WIS of -0.42, for instance, means a model whose WIS is on average $e^{0.42} \approx 1.5$ times smaller than the baseline's, i.e. about 1.5 times more accurate; a positive value of the same size would mean 1.5 times worse. One property of this metric constrains how its values should be read: because it is a fully-crossed pairwise comparison, each model's score depends on the set of models it is compared against, so the figures below apply to the six models in Fig 3 and shift if the comparison set changes. They are not directly comparable to a relative WIS computed elsewhere against a different set. Within this set, the deep network has the best mean skill (-0.42) but also the highest variability of any model except naive (0.24), the quantitative signature of its ordinary-season strength and outlier-season failure. The mechanistic model sits at the opposite corner: modest mean skill (-0.09) but the lowest variability of any model scored here (0.04; the seasonal-naive baseline is excluded from this comparison, since relative WIS is measured against it and its variability is zero by construction), because its bootstrap of historical seasons widens

Table 1: Dengue mean WIS by season (fold). Lower is better. Fold 2 contains the 2024 outbreak and is roughly ten times harder than any other season. Fold 4 (2025–26) is partially resolved and reported separately. Best value in each column is in bold.

Model	Fold 1 2022–23	Fold 2 2023–24	Fold 3 2024–25	Fold 4 2025–26
Ensemble (conformal)	337	3297	403	408
Ensemble (Vincentization)	338	3574	398	399
Ensemble (inverse-WIS)	**313**	3625	387	401
GRU (negative-binomial)	348	4131	324	542
GRU (pinball-trained)	361	3736	**321**	455
XGBoost	376	3238	484	468
Climatological	386	3588	516	309
Mechanistic	468	3371	551	314
Surge template	489	**3118**	511	329
Seasonal-naive	500	3848	581	293
Naive	726	3833	1056	**266**

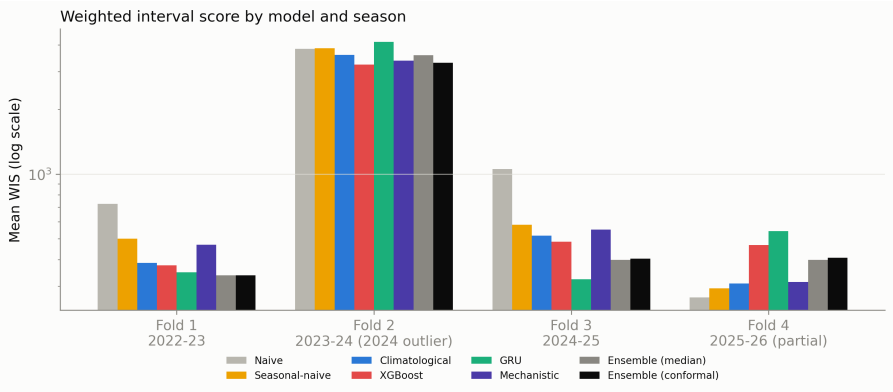


Figure 2: Mean WIS by model and season (log scale). Model rankings reorder across seasons. The deep network is the best single model in ordinary seasons and the worst of all on the 2024 outlier; the ensemble avoids every model's worst-season failure. The conformal-recalibrated ensemble (Section 2.3) is shown alongside its members. For legibility the figure plots the eight models of the deployed comparison; Table 1 adds the two extended-set models of Section 2.8.

intervals in proportion to historical variability rather than assuming a fixed noise scale. Both ensembles trade part of the deep network's mean skill for markedly better consistency (std 0.15 and 0.13 for the median and conformal ensembles), which is the quantitative counterpart of the qualitative claim above: an ensemble is not chosen for being the single best model in any one season, but for never being the worst.

2.3. An ensemble is the most robust model, and conformal recalibration makes it the most accurate on pooled WIS

Because different models win different seasons, a combination that is never worst is attractive. An unweighted per-quantile median ensemble, Vincentization, meaning that at every quantile level (the median, the 2.5th percentile, and so on) we take the median of the three models' values at that level, of the climatological, gradient-boosted (XGBoost), and deep-learning models avoids the catastrophic single-season failure of its weakest member (WIS 3574 in the outlier season, far below the deep network's 4131) but, once XGBoost replaced a weaker gradient-boosted alternative as the strongest single model (Section 4), no longer beats that single model outright on raw WIS alone

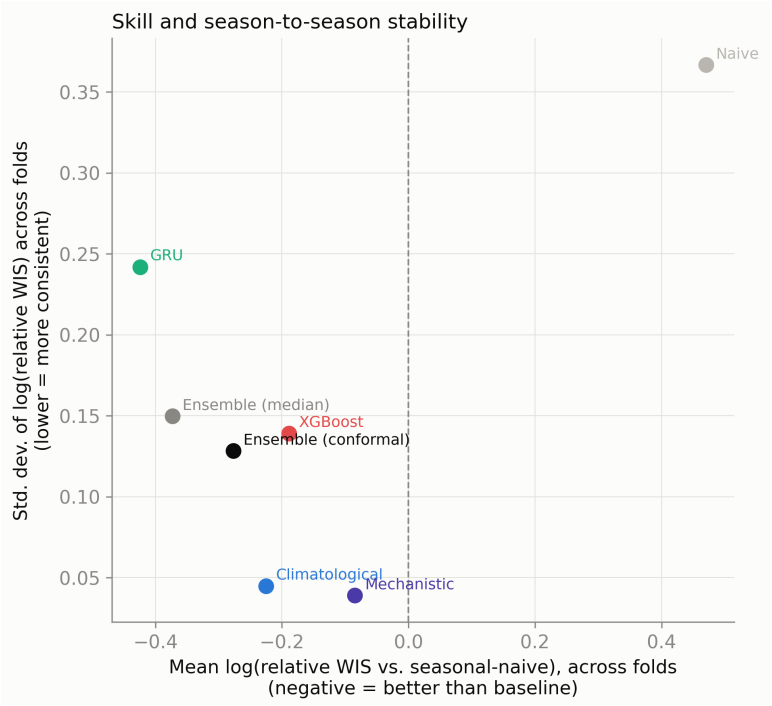


Figure 3: Skill and season-to-season stability are distinct axes. Each point is one model's mean log relative WIS against the seasonal-naive baseline across the four folds (x, negative is better) against the standard deviation of that same quantity (y, lower is more consistent). The deep network has the best mean skill but is among the least stable; the mechanistic model is the most stable but has only modest mean skill; the ensembles occupy an intermediate position that no single member reaches.

(1234 versus 1190 for XGBoost; Table 2). Its interval calibration is close to nominal but slightly narrow.

We improve on it with a conformal recalibration of the ensemble intervals, a distinct step from the conformalized quantile regression (CQR) used inside the gradient-boosted model (Section 4.6 gives the full procedure): both stretch prediction intervals to fix them being too narrow, but CQR calibrates one model's own intervals during fitting, whereas this step recalibrates the already-combined ensemble's intervals afterward. Calibration factors that make each central interval reach its nominal coverage are estimated on a single tuning season (fold 1) and applied to all seasons. The estimated factors are gentle and concentrate on the tails: 1.08, 1.04, 1.19, and 1.86 for the 50, 80, 90, and 95 percent intervals. Concretely, a raw 95 percent interval of, say, [40, 160] around a median of 100 becomes approximately [−12, 212], clipped to [0, 212] for non-negativity, after the 1.86 factor: the median is never touched, only the interval stretches, and it stretches only where it needs to (the 80 percent interval above needed only a negligible correction). This step lifts the ensemble to the top of the table rather than leaving it a hedge against any one member's worst case: the recalibrated ensemble has the lowest mean WIS of any model we evaluated (1162, against 1234 for the plain ensemble and 1190 for the best of the three deployed families, XGBoost; normalized WIS 0.561 versus 0.595) and is better calibrated at the tails (empirical coverage 46/74/87/94 versus 43/72/82/86), as shown in Fig 4. That lead is narrow at the top: the surge template (Section 2.8) reaches 1168, and the 6-unit gap between them is not statistically significant (Section 2.4). What separates them is where each earns its score, and we return to this in Section 2.8. Out of sample, on the three seasons not used for calibration, the reduction is 6.2 percent. The gain is concentrated in the outbreak season (fold 2 WIS 3574 to 3297) at a small

Table 2: Overall dengue backtest. Averaged over four seasons and 26 states. Mean WIS is magnitude-weighted; normalized WIS (sum of WIS divided by sum of cases) is the challenge's headline metric and is reported for all folds and excluding the 2024 outlier. Coverage nearer the nominal 50/80/90/95 is better. Best value in each column is in bold. The surge template and the pinball-trained GRU come from the extended set of Section 2.8, and the size-aware ensemble from Section 2.14; S5 Table lists all 25 models and variants evaluated, including those not retained.

Model	**WIS**	**nWIS**	**nWIS**	**Coverage (%)**			
		all	**ex-2024**	50	80	90	95
Ensemble (conformal)	**1162**	**0.561**	0.375	46	74	87	94
Surge template	1168	0.564	0.447	64	88	93	96
Ensemble (size-aware)	1173	0.566	0.368	42	72	82	87
XGBoost	1190	0.575	0.434	48	79	88	92
Ensemble (Vincentization)	1234	0.595	0.371	43	72	82	86
Ensemble (inverse-WIS)	1238	0.598	**0.358**	44	74	84	89
Mechanistic	1238	0.598	0.450	55	80	86	90
Climatological	1264	0.610	0.407	48	74	83	86
GRU (pinball-trained)	1274	0.615	0.366	33	56	70	78
LightGBM	1301	0.628	0.549	48	80	89	93
Seasonal-naive	1379	0.666	0.468	43	78	87	92
GRU (negative-binomial)	1394	0.673	0.385	27	50	60	67
Naive	1557	0.752	0.713	41	73	81	87

cost in ordinary seasons, so the recalibration behaves as asymmetric insurance: it widens the outer tails just enough to blunt catastrophic under-prediction in an extreme year while barely affecting normal seasons. Recalibrating the ensemble output outperforms recalibrating any single member, because the per-quantile median across members otherwise dilutes a single member's widening. The same factors cost skill when every state counts equally; a size-aware variant (Section 2.14) gives up under 1 percent of this mean-WIS lead to recover it.

2.4. The key differences are statistically significant

The absolute skill of each model carries wide uncertainty. A block bootstrap, repeatedly resampling whole state-season units (with replacement) and recomputing normalized WIS on each resample to see how much the result varies, over the 104 state-season units gives a 95 percent confidence interval (CI) on the conformal ensemble's normalized WIS of $[0.423, 0.639]$, and the intervals of the leading models overlap (Fig 5A), a direct consequence of the outbreak season inflating the variance of any magnitude-weighted score. The comparisons that matter are nonetheless decisive, because the models are strongly correlated across units, meaning they tend to do well or badly on the same state-seasons, so a paired analysis that looks at each model's difference from another on the very same units cancels that shared, season-driven variance instead of drowning in it (Fig 5B). The conformal recalibration improves on the median ensemble by a mean 71.3 WIS (95 percent CI $[56.9, 87.8]$) and on XGBoost by 27.9 ($[8.8, 47.3]$); on ordinary seasons the deep network beats XGBoost by 48.9 ($[14.9, 87.4]$); XGBoost itself beats the alternative gradient-boosted implementation we also tested, LightGBM, by 111.1 ($[81.6, 143.0]$); and the conformal ensemble beats the naive baseline by 395.1 ($[338.8, 457.3]$). All five intervals exclude zero. The lesson is methodological as much as substantive: in a record with one extreme season, marginal skill intervals are wide, but paired comparisons remain informative and are the sound basis for model selection.

The same test also bounds what we can claim at the top of the table. Against the surge

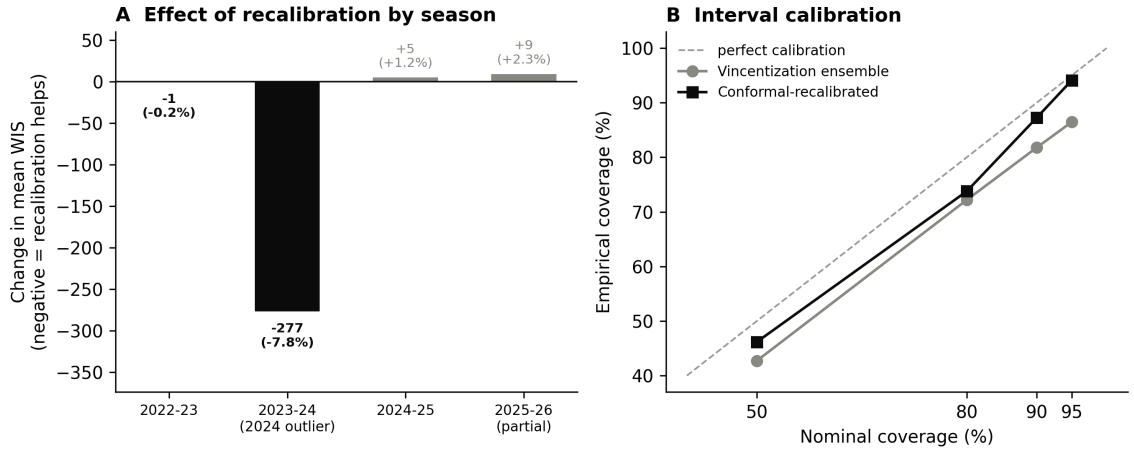


Figure 4: Effect of conformal recalibration on the dengue ensemble. (A) Change in mean WIS per season, conformal-recalibrated minus Vincentization ensemble, so negative means recalibration helps. The asymmetry is the point: a gain of 277 WIS units in the 2024 outlier season against costs of 5 and 9 units in the two ordinary seasons that follow it, roughly thirty to fifty times smaller. (B) Interval calibration: empirical versus nominal coverage. Recalibration moves the outer intervals (90, 95) toward the diagonal.

template, the best-scoring single model on mean WIS (Section 2.8), the conformal ensemble's advantage over all four seasons is 5.9 WIS with a 95 percent interval of $[-26.2, 37.4]$, which includes zero: on raw mean WIS the two are statistically indistinguishable, because the outbreak season the surge template handles well dominates that average. Restricted to the three ordinary seasons, the ensemble beats it by 73.9 ($[54.5, 91.7]$), an interval well clear of zero. We therefore do not claim the ensemble is significantly the most accurate model in raw aggregate. We claim something narrower and better supported: it is the most accurate in the seasons that recur, it is never the worst in any season, and it is the only model near the top of the table that also passes the tuning-fold check used to select it.

2.5. The winner depends on the evaluation metric

The overall mean WIS in Table 2 is dominated by the largest states and by the outbreak season, both of which contribute large absolute scores. The challenge's headline metric is instead a normalized WIS, the sum of WIS divided by the sum of observed cases, which is scale-free across regions. Under this metric the ordering changes in a revealing way. The conformal ensemble is best over all folds (normalized WIS 0.561), consistent with the mean-WIS ranking. When the 2024 outlier is excluded to isolate ordinary-season skill, the deep network rises from worst of six leading models by mean WIS to third-best by normalized WIS (0.385), and the plain median ensemble overtakes the conformal ensemble for the single best spot on this metric (0.371 versus 0.375), while the gradient-boosted model (XGBoost) falls from second-best to fifth of six (0.434) and the mechanistic model falls from fourth to last (0.450) (Fig 6). The reversal is real but milder for the gradient-boosted model than it was for the LightGBM configuration we tested first, which fell all the way to the weakest non-baseline model in Table 2 under this metric (0.549, still its value there since LightGBM itself was not adopted): XGBoost's more balanced profile across seasons (Section 2.2) narrows, but does not eliminate, the metric-dependent reordering that motivates reporting both metrics. The disagreement is not a paradox. Consider two toy states, one with 10,000 weekly cases and one with 100: a model that misses the large state's count by 1,000 and

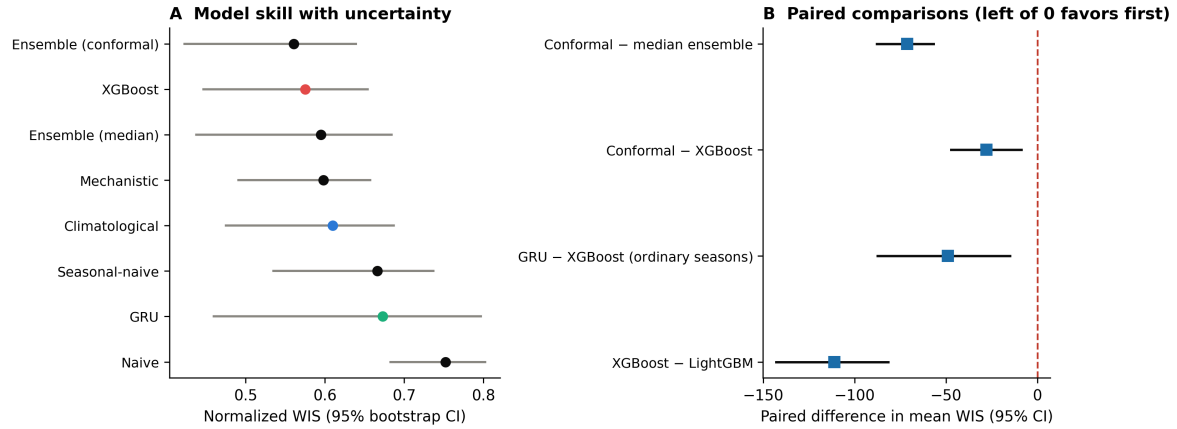


Figure 5: Statistical significance of the model comparisons. (A) Normalized WIS with 95 percent block-bootstrap intervals over state-season units; the leading models' intervals overlap because the 2024 outbreak inflates the variance of absolute skill. (B) Paired differences in mean WIS with 95 percent intervals; all four comparisons lie left of zero, so the conformal ensemble significantly beats the median ensemble and XGBoost, the deep network significantly beats XGBoost on ordinary seasons, and XGBoost significantly beats the alternative gradient-boosted implementation we also tested, LightGBM.

the small state's by 50 has a much larger magnitude-weighted error, dominated by the 1,000, than a metric that instead scores each state's error relative to its own size (a 10 percent miss on the large state versus a 50 percent miss on the small one, so a scale-free metric would actually rate the small state's forecast as worse). A metric that rewards accuracy on a few very large states favors different models than one that rewards accuracy on the many small states, and the 2024 season inflates the mean for whichever model handles it least badly. We therefore report both magnitude- and unit-weighted skill, and we caution that an epidemic-forecasting leaderboard reporting a single aggregate can mislead. The cross-state skew has a clear geographic structure (Fig 7): forecast difficulty, measured by normalized WIS, is greatest in the dense, high-incidence southeastern states and lowest in the low-incidence interior, so magnitude-weighted skill is determined by a few states.

2.6. Calibration and its correction

Raw gradient-boosted and deep-learning quantile models are over-confident (Fig 8): their nominal coverage, the fraction of outcomes an interval is supposed to contain, e.g. 90 percent for the 90 percent interval, is higher than their empirical coverage, the fraction it actually does contain. The GRU's nominal 50 percent interval only actually contains the truth 27 percent of the time, and its nominal 95 percent interval only 67 percent of the time, and it is under-covered in every season rather than only in the outbreak year, which identifies the problem as systematic over-confidence rather than a regime-specific failure. Conformalized quantile regression restores the gradient-boosted model (XGBoost) to near-nominal calibration (48/79/88/92) at negligible WIS cost. The same correction applied to the deep network trades away sharpness, how narrow and therefore informative its intervals are, on the seasons where it is already accurate, so calibration and sharpness trade off at the level of a single model. The per-quantile median of the ensemble provides a structural calibration improvement at no cost, and the conformal recalibration of Section 2.3 completes it.

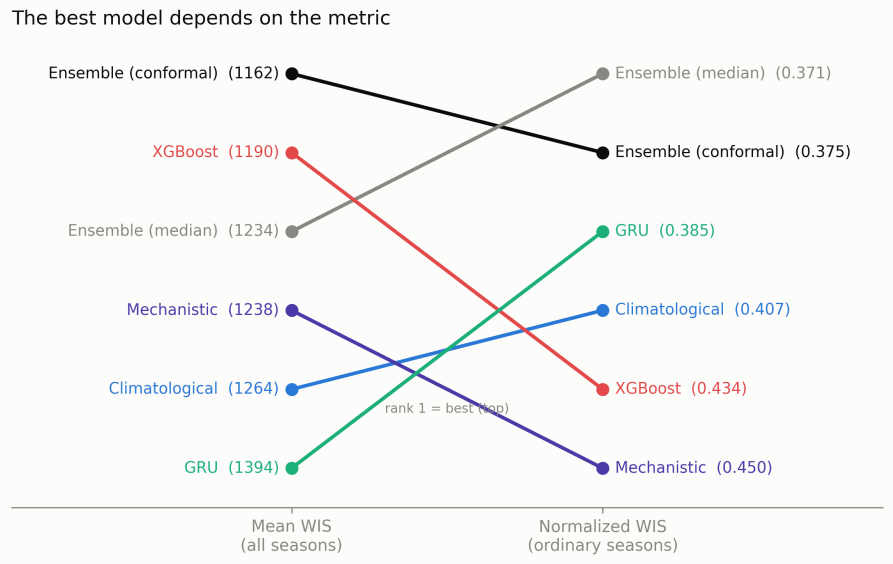


Figure 6: The best model depends on the metric. Model rank by magnitude-weighted mean WIS over all seasons (left) and by normalized WIS over the ordinary seasons with the 2024 outlier excluded (right); rank 1 is best, among these six models. The rankings reorder: the deep network rises from last by mean WIS to third on ordinary seasons, the gradient-boosted model (XGBoost) falls from second to fifth, the mechanistic model falls from fourth to last, and the plain median ensemble overtakes the conformal ensemble for the top spot. Values in parentheses are mean WIS (left) and normalized WIS (right).

2.7. Skill varies with horizon, and the outbreak failure is under-prediction

Because the 15-week reporting gap places every target 16 to 67 weeks ahead, we stratified skill by lead time on the ordinary seasons (Fig 9A). The deep network's advantage is horizon-dependent: it is the strongest of the four models stratified here at short lead times (normalized WIS 0.255 at 16 to 27 weeks, bettered only by its pinball-trained variant at 0.227, Section 2.8) but degrades at the longest horizons (0.388 at 52 to 67 weeks), where both the conformal ensemble (0.338) and the gradient-boosted model (0.349) overtake it. The conformal ensemble is the most stable model across horizons and is never worst, which mirrors its stability across seasons. WIS decomposes exactly into three non-negative components (Section 4.4): dispersion (a penalty for interval width, present even when the forecast is centered correctly), under-prediction, and over-prediction; decomposing it this way explains the outbreak-season failure directly (Fig 9B): in the 2024 season the ensemble's score is dominated by under-prediction (a mean of 2675 WIS units, against 621 for dispersion and 1 for over-prediction), whereas every other season is dominated by dispersion, i.e. is driven by ordinary interval width rather than the forecast being systematically off in either direction. The record epidemic was not mis-shaped but simply larger than any model would predict from pre-season information, which is why widening the upper tail, as the conformal recalibration does, is the effective remedy.

2.8. Nine further models, and what their failures identify

The five families above are the ones the deployed pipeline uses. We built and scored nine more, partly on our own initiative and partly by reimplementing techniques described by other teams in this challenge and its 2024 predecessor. S5 Table reports all of them on the identical backtest; four warrant discussion here because their outcomes locate where the predictable signal in this problem

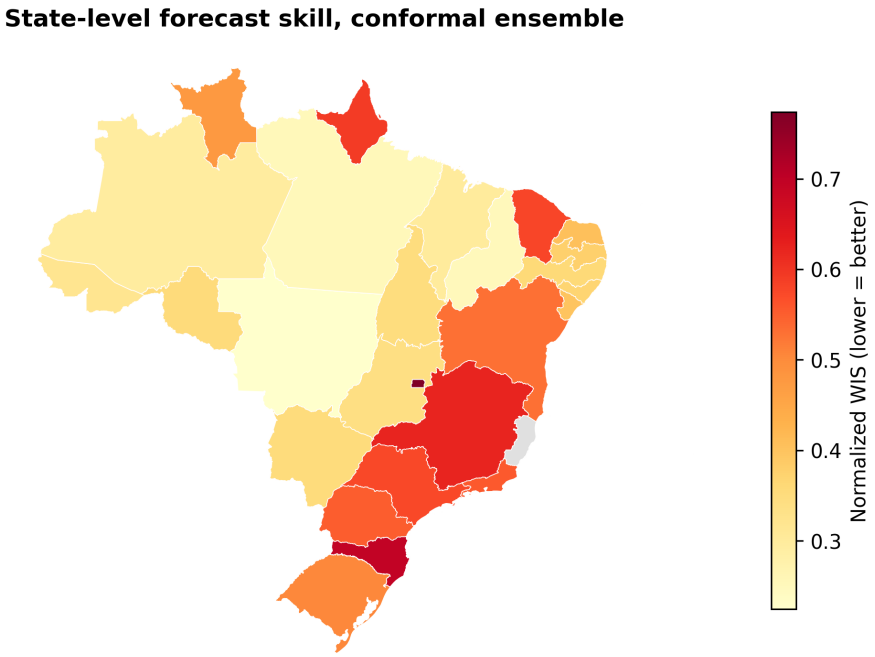


Figure 7: Geographic structure of forecast skill. Normalized WIS of the conformal ensemble by state (lower is better). Forecasting is hardest in the dense, high-incidence southeastern states and in the Federal District, and easiest in the low-incidence interior, which is why magnitude-weighted skill is dominated by a few states. Espírito Santo (gray) is excluded from the challenge.

is, and is not. Section 3.4 draws the common thread through them.

The **surge template** is the strongest single model we built by mean WIS (1168, against 1190 for XGBoost). It carries no covariates at all. It circularly shifts each historical season to align its peak, averages those into one normalized template shape, and forecasts by drawing a peak-week offset and a log-normal gain against that template. Its advantage is entirely a property of the outbreak season, where it scores 3118, the best in Table 1, because resampling whole historical seasons produces intervals that widen when the target is unlike the recent past. The two first-pass classical models make the same point far more starkly. Prophet records the best outbreak-season score of any model we built, 2812, and SARIMAX the second best, 2979, yet they finish last and 16th of 24 overall, because the very wide intervals that cover the 2024 surge are ruinous in the ordinary seasons (fold 3 WIS 2636 and 1464 respectively). Outbreak-season skill alone is not evidence of a good forecaster, and the model that best covers the worst season in the record can still be the worst model in the study. The same width is dead weight in an ordinary season, where it over-covers badly (64 percent empirical coverage at the nominal 50 percent level) and pays for that in dispersion: its normalized WIS excluding 2024 is 0.447, eighth of the twelve models in Table 2, and its tuning-fold normalized WIS is 0.444 against the deployed ensemble's 0.306. A model can therefore lead the raw aggregate while failing the criterion used to select models, which is the metric-disagreement result of Section 2.5 arriving through an independent route.

A **pinball-trained GRU**, identical in architecture to the deployed deep network but fitted to the quantile loss directly instead of through a negative-binomial likelihood, is significantly the better of the two as a standalone model (mean WIS 1274 versus 1394, paired difference 120.0, 95 percent CI $[91.0, 150.5]$), has the best ordinary-season normalized WIS of any single model (0.366),

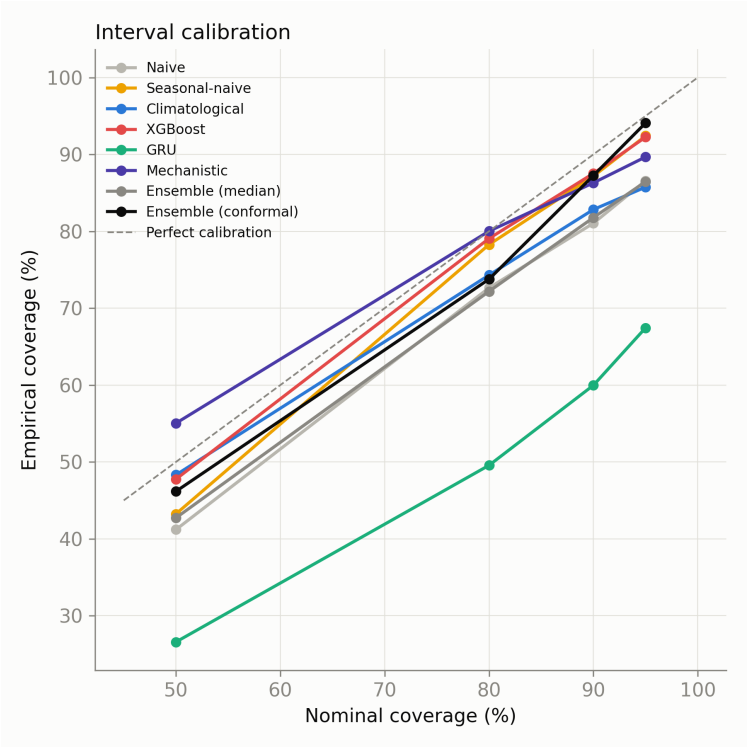


Figure 8: Interval calibration by model. Empirical versus nominal coverage; points on the diagonal are perfectly calibrated. The deep network is markedly overconfident; conformalized quantile regression corrects the gradient-boosted model, and the conformal-recalibrated ensemble tracks the diagonal at the outer intervals.

and is still overconfident, though less so (33/56/70/78 against 27/50/60/67). It is nonetheless the worse ensemble member: substituted for the negative-binomial variant it loses on every ensemble-level metric. Being a better model and being a better ensemble member are different properties, and the second depends on error correlation with the other members rather than on accuracy alone.

The **classical statistical family** deserves explicit mention because a Prophet model and a SARIMAX model from another team were named among the best-performing dengue state-level entries at this edition's validation round [36]. Ours were not. A first Prophet pass scored 2177 and a first SARIMAX pass 1419, both worse than seasonal-naive. Rebuilding each against the specific gaps we could identify, flat growth, real climate and ocean-index regressors, and a seasonally stratified residual bootstrap for Prophet, improved it to 1520, a 30 percent reduction that still left it well short. For SARIMAX, the order selected on one fold did not transfer to another regime (1752 when frozen, 1528 with a per-fold search), which is the same tuning-transfer failure documented in Section 2.9. We report these as negative results rather than omitting them: at a 16 to 67 week horizon with no recent case data near the target, the autoregressive structure both families rely on has little left to condition on.

Finally, **conditioning the surge template on climate** made it worse, not better (1406 versus 1168, degrading on every metric). Centring the forecast season's gain distribution on a regression against the pre-cutoff temperature anomaly added noise without signal, because an eight-week anomaly reading is a weak proxy for the climate of a season more than a year ahead. This is the same conclusion as the covariate ablation in Section 2.9, reached from the opposite

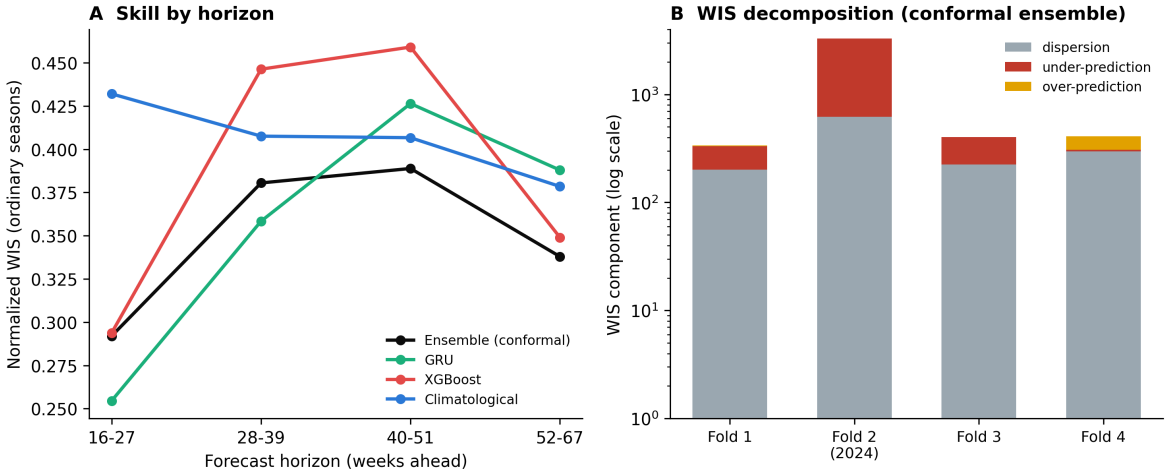


Figure 9: Robustness across horizons and anatomy of the outbreak-season failure. (A) Normalized WIS by forecast horizon on the ordinary seasons; the deep network leads at short lead times and fades at long ones, while the conformal ensemble is stable. (B) WIS decomposition of the conformal ensemble by season (log scale); the 2024 season is dominated by under-prediction, not by dispersion or over-prediction.

direction: the climate information that helps here is the forward-looking seasonal forecast and the persistent ocean indices, not a recent local observation.

2.9. Added complexity did not improve forecasts

Four interventions that plausibly should have helped did not (Table 3). First, adding observed-climate covariates to the gradient-boosted model, following the distributed-lag non-linear (DLNM) structure of the strongest model in the 2024 sprint, which lets a covariate's effect on cases be both delayed (this week's temperature can affect cases several weeks later, not just this week) and non-linear [1, 17], worsened backtest WIS by 3.3 percent (1299 to 1342 on the LightGBM configuration on which this test was originally run[1]). A feature-importance analysis showed why: the nine added covariates (minimum temperature, a derived standardized-precipitation drought index at one, three, and six month accumulations, multi-month temperature lags, thermal range, and rainy days) together carried only 5.3 percent of importance and ranked below the existing features. The climate signal the model actually uses is the seasonal climate forecast and the El

[1]This test and the hyperparameter-tuning test below were originally performed on the LightGBM-based pipeline before it was superseded by XGBoost (Section 4) and before a later data refresh. Re-verified on the current XGBoost-based pipeline and current data (2026-09-09): the climate-covariate finding replicates in the same direction (WIS 1166 → 1190 without vs. with the covariates, a 2.0 percent degradation from adding them, versus the original 3.3 percent). The hyperparameter-tuning finding does *not* replicate: on XGBoost, the config an internal fold-1 holdout preferred (shallower trees, `max_depth` 3 vs. the deployed default 5, with stronger leaf-size regularization) also won the true 4-fold backtest on every single fold (WIS 1190 → 1152, a genuine 3.2 percent improvement, not an overfit to the holdout) – the opposite of the LightGBM-era finding described in the main text below. We report both results rather than silently replacing the original figures, since the discrepancy is itself informative: whether an internal-holdout hyperparameter search transfers to the true backtest evidently depends on the specific model family and hyperparameter space searched, not a fixed property of this forecasting task. This XGBoost re-verification was checked at the ensemble level under this project's own fold-1 tuning discipline (the same check used to validate the original LightGBM-to-XGBoost swap) and did *not* pass: combined with climatological_quantile and gru_negbin and conformal-recalibrated, both the shallower-hyperparameter config alone and combined with dropping the climate covariates lose on fold 1 (WIS 347 and 346 respectively, vs. the deployed ensemble's 337) despite winning on every full-aggregate metric – the same overfitting-to-the-reporting-folds pattern this project has already caught in other candidate changes (Section 2.9). Dropping the climate covariates alone does pass fold 1 at the ensemble level (335 vs. 337) but is a mixed result elsewhere (worse raw WIS and normalized WIS all-folds, better normalized WIS excluding 2024) rather than a clean sweep. None of the three candidates were adopted; the deployed ensemble is unchanged.

Niño–Southern Oscillation indices, both already present and both forward-looking, rather than observed local weather, which is stale at the long horizons that dominate a full-season forecast. Second, a hierarchical, empirical-Bayes pooling of each state's seasonal climatology toward its macroregion, that is, borrowing statistical strength by pulling each state's estimate partway toward its region's average rather than trusting each state's own history entirely, worsened the ensemble from 1162 to 1185, because Brazilian states are too heterogeneous within a macroregion: shrinking a small state toward a region dominated by a larger, differently behaving state introduces bias. Third, a hyperparameter search that improved an internal time-block holdout, a held-out slice of recent training weeks used to tune hyperparameters before touching the real backtest, by 5 percent worsened the true backtest on the LightGBM configuration this was originally tested on, because an internal holdout does not necessarily represent forecasting a full future season from mid-year; conservative, a priori defaults generalized better in that specific test. A re-verification on the current XGBoost-based pipeline found the opposite as a standalone model (see footnote above): the holdout's preferred configuration also won the true backtest there. Checked further at the ensemble level, though, it lost on the same fold-1 tuning check that validated every other change reported here, so it was not adopted – a reminder that a standalone-model result and an ensemble-level result can disagree even when both are individually rigorous. Fourth, we tested the ensemble combination rule itself: the inaugural 2024 sprint's organizers combine models by log-linear (logarithmic) pooling of parametric predictive distributions [1, 2], a genuinely different rule from our per-quantile-median Vincentization – a precision-weighted combination of distributions rather than a per-quantile order statistic, which guarantees a unimodal result. Substituted for Vincentization (equal weights, then conformal-recalibrated identically to the deployed ensemble), it wins three of four folds outright, including fold 1, the only fold this project's discipline permits for tuning decisions (WIS 309 versus 337, an 8 percent improvement) and the ordinary-season metric (normalized WIS excluding 2024, 0.338 versus 0.375). By the letter of that discipline, this candidate passes and every previous one in this table failed. But it loses specifically on the 2024 outbreak fold (WIS 3630 versus 3297, a 10 percent regression, with 90 and 95 percent coverage both falling further from nominal, 64/79 versus 70/83): precision-weighted pooling is sharper in ordinary seasons but narrows the combined tails exactly where the widest tails matter most. We did not adopt it. The fold-1 rule exists to prevent tuning the ensemble to the specific reporting folds it is judged on, not to license a change that measurably weakens performance on the one season the entire ensemble-plus-conformal design is built to protect against; a rule applied mechanically here would have recommended trading away the outbreak-season robustness that motivates this paper's central design choice. Together these results locate much of the predictable signal in this problem, caution against complexity that is not validated on the true forecasting task, and show that even a validation discipline's own conclusions can be implementation-specific – and occasionally require judgment beyond the rule itself – rather than a fixed property of the problem.

2.10. Chikungunya is a more predictable, contrasting target

Brazil reported roughly 422,000 chikungunya cases in 2024, a cumulative incidence near 199 per 100,000, and the virus has recurred across the country since its 2014 introduction with a spatial pattern shaped by heterogeneous population immunity [31]. We forecast chikungunya on the same 26 states, harness, and seasons as dengue. Two differences stand out. First, the gradient-boosted

Table 3: Interventions that did not help, plus two that passed part of our validation discipline but still were not adopted (negative results throughout). Each was evaluated on the same backtest as the main comparison. †Originally tested on the LightGBM configuration before it was superseded by XGBoost and before a later data refresh; re-verified on the current XGBoost-based pipeline and current data (see footnote in text). The hyperparameter row's standalone-model finding reversed on re-verification, but checking it at the ensemble level (the same fold-1 discipline that validated every adopted change) found it should not be adopted either. ‡Passes the fold-1 tuning check outright but regresses specifically on the outbreak fold (see text); not adopted on substantive grounds despite passing the formal rule – none of these five rows changed the deployed model.

Intervention	Effect	Reason
Observed-climate covariates† (XGBoost)	WIS 1166 → 1190 (current); 1299 → 1342 (orig.)	5.3% importance; signal is forecast + ENSO, not observed weather
Hierarchical macroregion pooling (added to ensemble)	WIS 1162 → 1185	small-state bias; too correlated with climatological member
Hyperparameter tuning† (LightGBM, original)	internal holdout +5%, backtest worse	internal holdout misrepresented full-season forecasting
Hyperparameter tuning† (XGBoost)	standalone −3.2%; ensemble fold 1 worse	generalized alone, but not at ensemble level (see text)
Log-linear pooling‡ (vs. Vincentization)	fold 1 −8%; outbreak fold (fold 2) WIS +10%	sharper pooling narrows tails exactly where width matters most

Table 4: Chikungunya backtest. State-level, same harness and seasons as dengue. Gradient boosting alone is best.

Model	WIS	Cov. 50 (%)	Cov. 90 (%)
LightGBM	**79.1**	47	88
Ensemble (Vincentization)	89.5	55	87
Climatological	92.4	56	85
Seasonal-naive	95.1	48	88
Mechanistic	99.5	53	86
Naive	119.1	41	80

model alone is the best forecaster (WIS 79.1), ahead of the ensemble (89.5) and every other model (Table 4); it is best in every season (Table 5), so combining it with weaker models only dilutes it. Second, chikungunya is markedly more predictable than dengue relative to its scale: the 2024 season is only modestly harder than the others (mean WIS 120 versus a range of 33 to 81 across seasons), in contrast to the order-of-magnitude jump dengue shows in 2024. This matches reports that chikungunya dynamics in Brazil are more regular than dengue's [31], plausibly because immunity accumulates against a single antigenic type rather than interacting across four co-circulating dengue serotypes. The practical consequence is that the best model is disease-specific: a blended ensemble is right for dengue, a single well-calibrated learner for chikungunya, even within one surveillance system and pipeline.

2.11. City-level forecasting is substantially harder

The challenge also solicits municipal forecasts for 15 dengue and 10 chikungunya cities, an optional target we submitted with the climatological quantile model. City-level forecasting is much harder than state-level: the best model's normalized WIS is 0.60 for dengue cities and 0.87 for chikungunya cities (Table 6), against 0.56 and 0.62 at the state level. Interval coverage is also poorer (50 percent coverage of 39 percent for dengue cities), because many mid-sized municipalities have episodic, over-

Table 5: Chikungunya mean WIS by season. LightGBM is best in every season, unlike dengue, and the 2024 season (fold 2) is only modestly harder than the others.

Model	Fold 1	Fold 2	Fold 3	Fold 4
LightGBM	**81.2**	**120.2**	**69.2**	**33.4**
Ensemble (Vincentization)	88.7	141.1	73.8	41.5
Climatological	94.3	143.3	75.3	43.7
Seasonal-naive	92.8	152.3	77.1	44.6
Mechanistic	92.1	166.6	77.0	48.6
Naive	181.3	145.7	75.3	57.7

Table 6: City-level backtest (optional tracks; climatological quantile model submitted). WIS and normalized WIS averaged over the target cities and four seasons. For chikungunya cities, climatological and seasonal-naive are statistically indistinguishable (WIS 30.17 versus 30.17); no value is bolded there for that reason.

Disease	Model	WIS	nWIS	Cov. 50 (%)
Dengue (15 cities)	Climatological	**123.3**	**0.60**	39
	Seasonal-naive	131.8	0.64	30
	Naive	133.5	0.65	30
Chikungunya (10 cities)	Climatological	30.2	0.87	57
	Seasonal-naive	30.2	0.87	50
	Naive	34.4	1.00	33

dispersed series with long runs of zero counts that a per-week empirical quantile under-covers. A degenerate all-zero median is a concrete failure mode of these sparse series, which we addressed by switching the point forecast to a clamped predictive mean, the distribution's mean rather than its median, floored at zero so it can never predict a negative case count; a principled zero-inflated treatment is a clear direction for the municipal tracks. These results quantify the resolution penalty of forecasting at the level where interventions are delivered.

2.12. Onset, peak, and total burden are decision-relevant summaries that WIS does not directly show

WIS scores every week of the season equally, but health-system planning asks three narrower questions: when will the season start, how large will the peak be, and how many cases will there be in total. We compute, per state-season unit, the error in the first week each series reaches 10 percent of its own peak (onset), the timing and log-magnitude error at the true peak week (as in Section 2.2), and the log-ratio of predicted to observed total season cases (Table 7). No model wins on every one of these summaries, which is the same qualitative pattern as the main WIS results but on a different, more directly interpretable set of outputs. Among the models in Table 7, the gradient-boosted model (XGBoost) has the lowest onset-timing error (3.7 weeks mean absolute error, against 3.8 for the conformal ensemble), the climatological model has the lowest peak-timing error (4.5 weeks, against 4.9 for the ensemble), and the gradient-boosted model again gives the least biased total-season case count (a log-ratio of -0.01, i.e. predicting within 1 percent of the true total at the median unit, against 20 percent under-prediction for the ensemble and 56 percent for the mechanistic model). Every model under-predicts case counts at the true peak week specifically, by 42 to 76 percent at the median unit, more than it under-predicts the season

Table 7: Onset, peak, and total-burden accuracy, dengue state-level, four seasons and 26 states (104 units). Onset and peak-week errors are mean absolute error in weeks. Peak magnitude and total cases are the median log(predicted / observed) at the true peak week and over the full season respectively; 0 is unbiased, negative is under-prediction. Best value in each column is in bold.

Model	Onset MAE (weeks)	Peak-week MAE (weeks)	Peak magnitude (log-ratio)	Total cases (log-ratio)
Ensemble (conformal)	3.8	4.9	−0.71	−0.22
XGBoost	**3.7**	5.3	−0.55	−0.01
Climatological	4.2	**4.5**	−0.85	−0.32
GRU (deep learning)	4.1	5.0	−0.59	−0.12
Mechanistic	4.1	5.0	−1.43	−0.83
Seasonal-naive	4.2	4.8	−0.82	−0.30
Naive	4.9	17.4	−1.13	−0.14

Naive is the worst model by every measure in this table, including a 17-week peak-timing error; unlike in an earlier data vintage of this analysis, its total-cases figure no longer happens to look artificially competitive, which is itself a reminder that a single season-total number, like a single aggregate WIS, can be sensitive to incidental error cancellation and should not be read in isolation from the other columns.

total, which shows that the harder part of the forecasting problem is capturing acute intensity at the peak rather than accumulated burden over the season. The ensemble is a reasonable middle choice on every one of these summaries without being the best on any single one, consistent with it being chosen for robustness (Section 2.2) rather than for leading any one metric.

2.13. The challenge organizers' independent evaluation agrees, and qualifies the recalibration

Every result above comes from our own harness. The challenge organizers scored all submitted validation forecasts independently [7], against their own reference model, a Bayesian model with no covariates, rather than our seasonal-naive baseline. Their score is the ratio R_v of a model's mean WIS to the reference's in one state and validation season, so $R_v < 1$ beats the reference. Our submitted dengue ensemble was one of five models that beat the reference in a stable fashion, with a geometric-mean R_v of 0.932 over all states, and it beat the reference in more states than any of the other four (20 of 26; they score 16 to 18). The five are close on mean skill: their geometric means span only 0.891 to 0.932. We reproduced this evaluation from the platform's own stored predictions for all 43 models registered to the 2026 challenge, scored with our WIS implementation. Our submission scores 0.933 and 20 states, matching the official figures. On the organizers' stability axis (the standard deviation of $\log R_v$ over state-season pairs) it scores 0.283, tied for the lowest of the five with the EMAp Prophet model (0.276; the other three, 0.339 to 0.536). Outside the stable group, EMAp's state-level SARIMAX model scores better on mean skill (0.853, 21 states) with a larger spread (0.492). For chikungunya, which the official summary does not tabulate in full, our LightGBM model scores 0.723 with 19 states below 1, third of 12 complete submissions behind two variants of EMAp's SARIMAX model (0.635 and 0.679), and tied with both for the most states. By season, the dengue submission scores 0.952, 0.944, 0.832 and 1.014 over 2022–23 to 2025–26, so it loses narrowly to the reference only in the partially resolved fourth season. The six states where its multi-season R_v exceeds 1 are GO, RR, AM, MA, PI and RS, mostly in the North and Northeast; RS reflects one season ($R_v = 2.79$ in 2025–26).

The same per-state metric qualifies the conformal result of Section 2.3. The validation forecast

the organizers scored carries the Vincentization ensemble's intervals; the recalibration was applied to the forecast-phase submission. Rescoring our own backtest against the organizers' reference, the Vincentization ensemble scores 0.926 and the conformal ensemble 0.989, worse in all four seasons. The two results do not conflict. Pooled mean WIS is dominated by the highest-burden states, where widening the outer tails pays for itself in 2024; a geometric mean of per-state ratios weights every state equally, and a single multiplicative factor (1.86 at the 95 percent level) over-widens the intervals of low-burden states. Which recalibration is better therefore depends on whether the forecast user's loss is national (pooled cases) or per state, the same metric dependence as Section 2.5, and it motivates a recalibration that serves both.

2.14. A size-aware recalibration serves both metrics

The size-aware recalibration replaces the single global factor with one that grows with a state's burden (Section 4.7). Each state's whole predictive distribution is rescaled about its median, in log space, by $s_u = a + b\, z_u$, where z_u is the state's standardized log mean weekly case count before the tuning season. Both parameters are fitted on the 2022–23 tuning season alone. They minimize the mean over states of the log of mean WIS, which has the same minimizer as the organizers' geometric-mean R_v against any fixed reference, subject to a pooled mean WIS no worse than the unscaled ensemble's. The rule selects $a = 1.00$ and $b = 0.10$, which sharpens the intervals by up to 23 percent in the smallest states (RR, AP) and widens them by up to 20 percent in the largest (SP, MG). Against the organizers' reference, the size-aware ensemble scores 0.916 with 20 states below 1 (0.926 for the plain ensemble, 0.989 for the conformal one), and 0.931, 0.934, 0.759 and 1.065 by season. Its pooled mean WIS is 1173, so it recovers 85 percent of the conformal ensemble's gain over the plain ensemble (1234 to 1162); its normalized WIS is 0.566 over all seasons and 0.368 without 2024 (Table 2). In paired comparisons it beats the plain ensemble on mean WIS by 60.6 (95 percent CI $[46.0, 77.2]$), and it beats the conformal ensemble on R_v by a factor of 0.926 ($[0.905, 0.947]$, bootstrapped over the 104 state-season pairs). It loses to the conformal ensemble on mean WIS by 10.7 ($[5.7, 16.2]$), under 1 percent, and its R_v gain over the plain ensemble (a factor of 0.988, $[0.974, 1.001]$) is not significant. It is thus not a strict improvement on either parent. It is, however, the only one of the three within 1 percent of the best on both metrics: the conformal ensemble is 8 percent worse than it on R_v, and the plain ensemble 5 percent worse on mean WIS. The cost is tail calibration. Because the rule sharpens small states, its coverage (42/72/82/87 percent) stays as narrow as the plain ensemble's and does not reach the conformal ensemble's 46/74/87/94.

Two simpler alternatives did worse. A single sharpening factor for every state, tuned on the same season (0.75), reaches an R_v of 0.885 but raises pooled mean WIS to 1355 and cuts 90 percent coverage to 72 percent. Separate factors for each state, each chosen on the seasons before it, overfit (0.933). For chikungunya, the same rule applied to the LightGBM forecast selects no change ($a = 1$, $b = 0$): on the tuning season, no burden-dependent rescaling improves the per-state score without raising pooled WIS. A uniform 20 percent sharpening would lower the chikungunya R_v from 0.715 to 0.644, but it would raise pooled mean WIS from 79 to 83, which is the same trade-off.

Two caveats limit these results. First, the rule was devised in November 2026, after the organizers' results had shown where our intervals lose. Its constraint and parameter grid were

fixed after we had seen exploratory scores on all four seasons, and only its two parameters were fitted blind to the 2023–24 to 2025–26 seasons, so the backtest gain is not a clean out-of-sample estimate. The 2026–27 season provides that test. We froze a size-aware version of the submitted forecast, with identical medians and size-aware intervals, generated from data through EW25 2026, and committed it to the project repository on 9 November 2026, before examining any 2026–27 observation. That date falls four weeks into the target window, so the strict comparison scores only the weeks after it. Both versions will be scored when the season resolves. Second, the rule does not close the gap to the leading submissions, because their advantage is mostly in the median. EMAp's SARIMAX model has 90 percent intervals about a quarter narrower than ours at the same 83 percent coverage. That is possible only because its median is more accurate: its absolute error is 9 percent lower than ours, both measured relative to the reference's WIS. Rescaling our intervals cannot reproduce a better median.

2.15. Most of the 2026–27 forecast's excess over the official ensemble is El Niño

Only three teams fed the 2026–27 climate outlook into their forecasts, and the organizers singled ours out as amplifying the developing El Niño most strongly, two to three times their ensemble in PR, SP, MG, the Center-West and BA; the official ensemble itself carries no El Niño signal [7]. We quantified that effect by re-predicting the submitted forecast with its climate-outlook inputs neutralized at prediction time, leaving every fitted model unchanged. The regenerated forecast reproduces the submission exactly (1,088,267 cases, summing state medians over the 26 states). Only the gradient-boosted member sees these inputs. Setting the ENSO index to neutral cuts that member's season total from 1.79 to 0.78 million and the ensemble's to 0.77 million, so ENSO alone raises our national forecast by 42 percent. Neutralizing the IOD and PDO indices and removing the ECMWF seasonal forecast as well lowers it only to 0.75 million: ENSO is nearly the entire climate effect. Against the official ensemble (0.62 million on the same basis) our forecast falls from 1.74 to 1.23 times when ENSO is neutral, so ENSO accounts for 69 percent of the excess. In the flagged states together the ratio falls from 1.86 to 1.29. The ENSO effect is strongest in the South and Southeast (a factor of 2.9 in RS, 2.4 in SC, 2.0 in MG and DF, 1.65 in PR) and near zero in the North (at most 1.17 in AM). The remaining excess comes from the GRU member, which sees no ENSO input but forecasts high totals in the same states from their recent incidence. For context, the organizers' covariate-free reference model forecasts about 1.02 million cases over the same states, close to our deployed total. For chikungunya, neutral ENSO lowers our LightGBM forecast by 16 percent (60,488 to 51,067 cases), against 53,261 for the official ensemble.

How far to trust the amplification depends on how many analogues the model learned it from. At the forecast origin (EW25 2026) the ENSO index had risen from 0.34 to 1.10 in seven weeks. The only training origin with a comparable June value is 2015 (1.55), which preceded the large 2015–16 season; June 2023, before the record 2024 outbreak, was still 0.36. The model's El Niño response therefore rests on essentially one prior event, and the 2026–27 season is the first out-of-sample test of it. Because the median ensemble takes the middle member at each quantile, the gradient-boosted member's 2.3-fold ENSO response reaches the ensemble only partly, which is the same never-worst protection that motivated combining models in Section 2.3.

3. Discussion

3.1. Robustness under regime shift

The most consequential finding is that model rankings are season-dependent and can invert in an outbreak year. A model chosen for its average score can be the least reliable in the season for which forecasts matter most, as the deep network is here. Two responses follow. One is to combine models, so that no single failure mode determines the forecast; the ensemble is never worst in any season and is best overall once recalibrated. The other is to prefer models whose uncertainty widens under distribution shift, as the mechanistic model's bootstrap-of-historical-seasons intervals do. These observations align with evidence from collaborative forecast hubs, where combined forecasts are consistently competitive with or better than their members [19–21]. The same holds for dengue specifically: across more than 180 locations worldwide, ensembles ranked consistently among the top performers at one-to-three-month horizons, including where individual models faltered [3]. Our result extends that to a full-season horizon and identifies the mechanism in this setting: the ensemble is never worst because its members fail in different seasons, and conformal recalibration then converts that robustness into the best overall score. The challenge exists to give Brazil's Ministry of Health a pre-season view of what the coming year may bring [1], and that use carries a concrete recommendation from these results: prefer a recalibrated ensemble to the single best-scoring model on a leaderboard. The season in which a leaderboard-topping model fails is, by construction, the season with no precedent in the training record, which is exactly the season a preparedness decision depends on most.

3.2. Metric choice is a reporting-standards issue

The identity of the best model changes with the aggregation of skill across a highly skewed set of states. This is not specific to our models or to Brazil. Any national forecasting evaluation with strong heterogeneity in population and incidence faces the same tension between magnitude-weighted and unit-weighted skill. Reporting a single aggregate obscures it. We recommend reporting both, as the interval-score and forecast-hub literature increasingly does [9], and we treat the normalized WIS the challenge adopts as the headline while retaining magnitude-weighted mean WIS and per-unit relative WIS alongside it.

3.3. Simple methods and the limits of complexity

Deep learning did not dominate, and simple methods, a climatological quantile model and a bootstrap of historical seasons, were hard to beat. This is consistent with forecasting-competition evidence that in small-sample, few-series regimes, parsimonious methods and combinations are strong [6, 26]. The pattern recurs one sprint earlier: in IMDC24, a Bayesian baseline model using no climate covariates at all was among the most consistently competitive submissions across states and seasons [1], echoing our own finding that a covariate-free climatological baseline and a historical-trajectory bootstrap are hard to beat with more heavily engineered alternatives. It recurs again against an architecturally close competitor: on monthly dengue counts in Freetown, Sierra Leone, a parsimonious autoregressive count model (INGARCH with a negative-binomial observation model) beat a bidirectional LSTM with a negative-binomial head, the closest published

analogue of our own GRU, on distributional accuracy, with the deep model preferred only where tail reliability mattered most [5]. Three independent settings, two continents, and two surveillance systems give the same answer, which makes it a property of the forecasting regime rather than of any one pipeline. The same predecessor sprint offers an independent caution about tuning ensembles on limited past data: its CRPS-weighted logarithmic-pooling ensemble, with weights fit to 2023–2024 performance, badly overestimated the true 2025 outbreak in three of five states, because the models it weighted most heavily had performed well on those past seasons but poorly on the actual target season [1] – the same overfitting-to-past-folds failure mode our fold-1 tuning discipline (Section 2.9) exists to catch, now observed independently in another team's ensemble on a different edition of the same challenge. Our negative results sharpen the point. The covariate that helped the strongest 2024-sprint model, climate acting through a distributed-lag non-linear response (DLNM, Section 2.9 above) [17, 18], did not help our gradient-boosted forecaster, because that structure requires the regularized, partially pooled Bayesian formulation in which it was developed, whereas an unregularized tree over the same inputs overfits low-signal dimensions. The forecast-relevant climate information in our pipeline is the seasonal forecast and large-scale oscillation indices, not observed local weather. This clarifies where future modeling effort should go: toward a Bayesian spatiotemporal member that can use climate through an exposure-lag-response surface, rather than toward more covariates in a boosted model.

3.4. What separated the models that worked from the ones that did not

Across the 25 models and variants we evaluated, spanning statistical, machine-learning, deep-learning, mechanistic, and template-based approaches, performance tracked one property more closely than model class: how each model generates its prediction intervals, and therefore how those intervals behave when the target season is unlike the training record. WIS decomposes exactly into a dispersion penalty, paid for interval width whether or not it is needed, and directional penalties, paid when the observation falls outside the interval (Section 4.4). Which of those two costs a model incurs is close to a design decision.

Models that build intervals by resampling whole historical seasons, the semi-mechanistic bootstrap and the surge template, produce width that scales with historical variability. They pay dispersion continuously and are rarely caught out: on the 2024 outlier the surge template ranks third and the mechanistic model ninth of the twenty-one models for which we have per-season scores (3118 and 3371), while neither is competitive in ordinary seasons, where that width is simply a cost. Models that derive intervals from a fitted parametric distribution, principally the two deep networks, pay almost no dispersion and are sharp when the season is ordinary, which is why the two of them hold the best ordinary-season normalized WIS of any single model, 0.366 for the pinball-trained variant and 0.385 for the deployed one. The same sharpness is catastrophic when the observation lands far outside: the negative-binomial GRU covers 27 percent of observations at its nominal 50 percent level and records the worst outlier-season score of any model, worse than a naive baseline. This is a trade-off, not a ranking, and it explains why a single leaderboard position is a poor summary of a forecasting model.

Three further patterns are worth stating because each was learned from a failure. First, covariates helped only when they were forward-looking. Observed local climate, whether added to the boosted model (Section 2.9) or used to condition the surge template's gain (Section 2.8), degraded

forecasts in both cases; the seasonal climate forecast and the persistent ocean indices, which carry information about the target window rather than the recent past, are retained in every deployed model. At a 16 to 67 week horizon this is close to a definitional point: a covariate observed before the cutoff has decayed to noise by the target week unless the quantity itself persists. Second, autoregressive structure has little to condition on at this horizon, which is the common thread linking the Prophet and SARIMAX results to the general weakness of recent-lag features. These families are strong at the one-to-three-month horizons where comparable studies deploy them [3, 5] and weak here, so the appropriate conclusion is about horizon, not about the families. Third, every intervention that improved a model in isolation had to be re-checked in combination, and two of them reversed: the pinball-trained GRU is significantly better standalone yet a worse ensemble member, and the shallower boosted configuration won the standalone backtest yet lost at the ensemble level. Ensemble value is a function of error correlation, which standalone accuracy does not measure.

3.5. Conformal recalibration as transferable insurance

The conformal recalibration is simple, scale-free, and cheap, and it improves both accuracy and calibration. It connects to conformalized quantile regression [11] and to the broader literature on adaptive conformal methods for distribution shift [12, 33], which similarly recalibrate intervals in response to a changing data-generating process rather than assuming a fixed noise scale, though those methods update continuously as new data arrive whereas ours is calibrated once, on a single tuning season. Because it widens mainly the outer tails, it protects against the failure mode that dominates the outbreak season without materially degrading ordinary-season sharpness. For an operational forecaster who cannot know in advance whether the coming season will be ordinary or extreme, this asymmetric protection is the right default when the loss is national. When each state's forecast matters equally, the single global factor over-widens low-burden states (Section 2.13). The version to transfer then has the factor depend on burden rather than on each state separately (Section 2.14). Separate per-state factors overfit, while a two-parameter burden rule keeps the per-state skill and gives up under 1 percent of the national gain.

3.6. Limitations

The fourth backtest season is only partially resolved in the available data and is reported separately; its numbers will firm up as the 2025–26 season completes. Municipality-to-state aggregation discards sub-state epidemic asynchrony that a municipal model could exploit. The study covers two diseases in one country and is retrospective; we submitted the recalibrated ensemble to the challenge's forecast phase (the true 2026–27 season, EW41 2026 through EW40 2027) ahead of its 2026-09-10 deadline, but a genuine prospective score awaits that season's resolution. The size-aware recalibration (Section 2.14) was designed after the validation results were known, so only that season can test it cleanly. Our model-family comparison is single-team; the organizers' cross-team evaluation (Section 2.13) places it against the other submissions but does not capture the full diversity of approaches a multi-team sprint would, though our 26-state, four-track design covers more geographic and epidemiological ground than the inaugural 2024 sprint, which evaluated five highly endemic states and explicitly left lower-incidence, more recently affected regions untested [1]; the conformal factors are calibrated on one season under the challenge's tuning-fold discipline

rather than by a fully online procedure. Finally, the 2023–24 outlier season we treat as a pure epidemiological surge may also reflect a data-quality confound: the 2024 season saw a substantial expansion of Oropouche fever, whose symptoms overlap with dengue's, and the inaugural sprint's organizers flag possible dengue-notification contamination from Oropouche misclassification as a contributing factor to that season's models failing so widely [1], a possibility our own pipeline cannot rule out with notification data alone.

3.7. Future work

The immediate next step is scoring the submitted forecast phase once the 2026–27 season resolves, which will provide a genuine prospective test of the recalibrated ensemble and, through the size-aware forecast we froze before the season, of the two recalibrations against each other. Beyond it, four directions follow from our results and from the challenge organizers' own stated plans. First, a Bayesian spatiotemporal or endemic-epidemic member with a distributed-lag non-linear climate response [34, 35], which our negative result suggests is the right home for climate covariates. Second, hierarchical reconciliation across state, city, and national forecasts for coherence. Third, municipal-level feature engineering for the optional city tracks. Fourth, the organizers report moving their own collective ensemble toward a copula-based approach that samples coherent full-trajectory epidemic curves rather than combining marginal per-week quantiles independently; our per-quantile median ensemble does not guarantee this, but our mechanistic model's bootstrap of complete historical trajectories already provides trajectory-level coherence by construction, which is a reason to prefer it, or a copula-linked resampling of it, over a purely per-week combination when coherent full-season trajectories rather than marginal weekly intervals are the target.

4. Materials and methods

4.1. Pipeline overview

Fig 10 summarizes the full pipeline end to end before the following subsections detail each stage. Raw weekly surveillance and covariate data (Section 4.2, below) are filtered to each fold's own training cutoff, which removes any observation dated after that cutoff before anything downstream can see it (Section 4.3). For the machine-learning families, the cutoff-filtered history is expanded into a synthetic-origin training panel: every historical week becomes a candidate forecast origin, paired with the realized value at horizons 1 to 67 weeks ahead, which supplies enough training examples for gradient boosting and the deep network despite only four official evaluation windows existing. Four model families are then fit independently per fold: baselines and the climatological quantile model, a gradient-boosted model with conformalized quantile regression, the GRU deep ensemble, and the semi-mechanistic trajectory bootstrap. We evaluated two gradient-boosting implementations on the identical panel, LightGBM [13] and XGBoost [14], and adopted XGBoost, which scored significantly better (Section 2.4; Fig 5). The climatological, XGBoost, and GRU outputs (not the mechanistic model, which is evaluated separately for comparison) are combined by per-quantile median (Vincentization) into an ensemble, which is then conformally recalibrated using widening factors estimated once on a designated tuning season (Section 4.6). The output at every stage is the same canonical shape, nine quantiles per state or city and week, and the final recalibrated quantiles are validated against the platform's submission schema (non-negative,

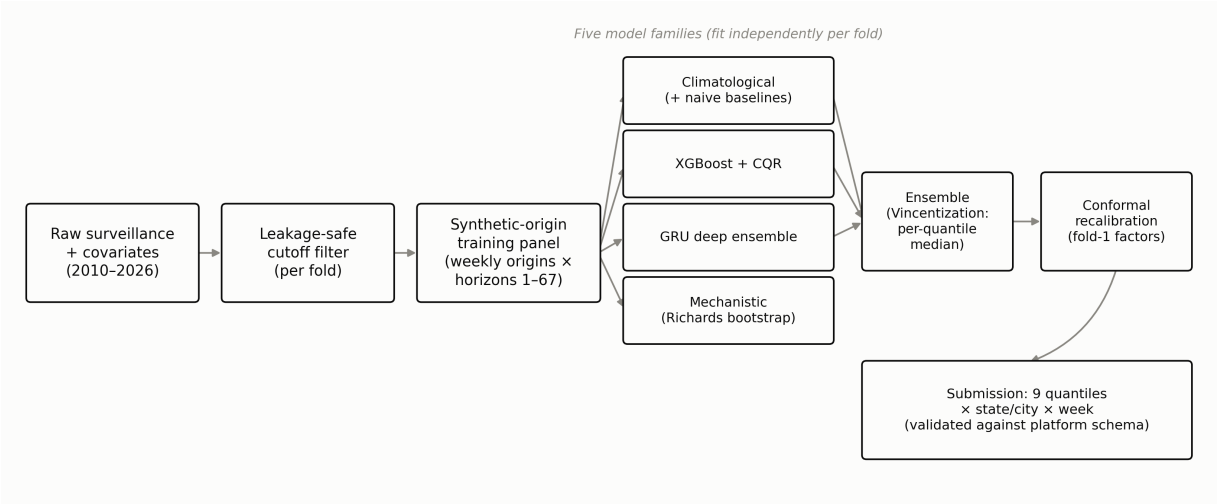


Figure 10: Pipeline overview. The dengue state-level pipeline, the most complex of the four tracks: cutoff-filtered data becomes a synthetic-origin training panel, four model families are fit independently, three of them (climatological, XGBoost, GRU) are combined by per-quantile median into an ensemble, and conformal recalibration produces the final submitted quantiles. The mechanistic model is evaluated for comparison but is not an ensemble member. Chikungunya-state and the city tracks (not shown) follow the same stages but submit a single family's output directly.

correctly nested, no date gaps) before being written out. Chikungunya-state and the two optional city tracks follow the same pipeline but submit a single family's output directly (LightGBM for chikungunya-state, unchanged from the dengue pipeline's original configuration, climatological for both city tracks; see Sections 2.10 and 2.11) rather than the three-member ensemble.

4.2. Data and targets

Weekly dengue and chikungunya case counts per municipality (2010–2026) are drawn from the Infodengue system, which harmonizes notifiable-disease surveillance [24], and aggregated to the 26 mandatory states, excluding Espírito Santo per the challenge. Covariates comprise weekly climate reanalysis (temperature, precipitation, humidity, pressure, thermal range, rainy days), a seasonal climate forecast, ocean-oscillation indices (El Niño–Southern Oscillation, Indian Ocean Dipole, Pacific Decadal Oscillation), yearly municipal population, and static climate-zone and biome covariates. The forecasting target is the weekly state case count over each season, submitted as the nine quantiles 0.025, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95, and 0.975.

4.3. Backtest design and leakage control

The four official seasons are derived at runtime from the challenge's train and target flags. A confirmed 15-week gap separates each training cutoff from the start of its evaluation window, simulating reporting delay. Two leakage disciplines are enforced centrally in the harness, not in individual models. All covariate access is filtered to dates at or before the training cutoff, and lag and rolling features are computed from the continuous cutoff-filtered history rather than from the gapped target window. Both are checked by automated regression tests, so a leak fails the test suite instead of silently inflating scores.

4.4. Scoring

Forecasts are scored by the weighted interval score,

$$\mathrm{WIS}(F, y) = \frac{1}{K + \frac{1}{2}} \Big[\tfrac{1}{2} \, |y - m| + \sum_{k=1}^{K} \frac{\alpha_k}{2} \, \mathrm{IS}_{\alpha_k}(\ell_k, u_k, y) \Big],$$

over the four required central intervals, where m is the predictive median, IS_α is the interval score at level $1 - \alpha$ (the interval's width plus a penalty, scaled by $2/\alpha$, for however far the observation falls outside it), and the weights reproduce the mean pinball loss over the nine required quantiles, the exact discrete approximation to the continuous ranked probability score [9, 10]. Concretely, for a simplified two-interval version of the score with median $m = 10$, a 50 percent interval $[6, 14]$, a 90 percent interval $[2, 20]$, and an observation $y = 15$: the median term is $\frac{1}{2}|15 - 10| = 2.5$; the 50 percent interval ($\alpha = 0.5$) is exceeded by 1 (since $15 > 14$), giving $\mathrm{IS}_{0.5} = (14 - 6) + \frac{2}{0.5}(15 - 14) = 12$, weighted by $\alpha/2 = 0.25$ to 3; the 90 percent interval ($\alpha = 0.1$) is not exceeded, giving $\mathrm{IS}_{0.1} = (20 - 2) = 18$, weighted by 0.05 to 0.9; so, with $K = 2$ intervals, $\mathrm{WIS} = (2.5 + 3 + 0.9)/(2 + 0.5) = 2.56$. Our implementation is verified against this hand-computed value, against an independent pinball-loss re-derivation, and against the challenge platform's own scorer. We also report empirical interval coverage, a decomposition of WIS into dispersion and directional (over- and under-prediction) penalties, a scale-free per-unit relative WIS, and the normalized WIS (sum of WIS divided by sum of observed cases) that the challenge adopts as its headline metric.

Model comparisons are tested by a block bootstrap over whole state-season units (Section 2.4) rather than by a Diebold-Mariano test [8], the more common choice for comparing two forecast series. The Diebold-Mariano statistic is built for a single long series of forecast errors with weak serial dependence. Our panel is 26 short state series that are cross-sectionally correlated, with one season an order of magnitude harder than the rest, so resampling whole state-season blocks preserves the dependence structure that an asymptotic test on pooled errors would assume away.

4.5. Models

Baselines. A naive model (last observed value with empirical horizon-difference quantiles), a seasonal-naive model (same-epidemiological-week history), and a climatological quantile model (direct order-statistic quantiles per state and epidemiological week over a five-week window).

Gradient boosting. Only four official evaluation windows exist per disease, far too few to fit a tree ensemble, so a gradient-boosted quantile model is instead fit on a synthetic-origin panel: within each fold's own cutoff-filtered history, every Sunday from roughly 2014 onward becomes a candidate forecast origin, and every (origin, horizon) pair from 1 to 67 weeks ahead becomes one training row, with origin-anchored lag, rolling, seasonal, climate, ocean, seasonal-forecast, and static features as inputs and the realized case count that many weeks later as the label. For example, one row might read: origin = the first Sunday of March 2021, horizon = 12 weeks, features computed using only data available on that March Sunday, label = the observed case count 12 weeks after it. This turns roughly 500 usable weekly origins per state, crossed with up to 67 horizons, into tens of thousands of training rows for a single fold, while evaluation still only ever happens at that fold's own four-times-a-year official target window. We evaluated two gradient-boosting implementations on this identical panel: LightGBM [13], one booster per quantile level,

and XGBoost [14], a single booster with a native multi-quantile objective covering all nine levels jointly. XGBoost scored significantly better both overall and on the outlier season (Sections 2.2 and 2.4) and is the version used in the main ensemble; LightGBM remains the gradient-boosted family used for the chikungunya-state track, on which it was not re-evaluated against XGBoost. Predictive intervals for both are calibrated by conformalized quantile regression (CQR) [11]: a held-out slice of the training panel is used to measure how much each quantile needs to be shifted so that it achieves its nominal coverage, and that shift is then applied to the model's raw quantile predictions, in the same spirit as, but computed independently from, the ensemble-level conformal recalibration of Section 4.6.

Deep learning. A global gated recurrent unit (GRU) network [15], a type of recurrent neural network designed to retain information over many time steps, with per-state embeddings and a negative-binomial output head (chosen over a Poisson head because case counts are more variable than a Poisson distribution allows), trained as a deep ensemble of independently seeded runs [16]; quantiles are analytic, meaning computed directly from the fitted negative-binomial distribution's formula rather than by simulation, and pooled across ensemble members by quantile averaging.

Semi-mechanistic. A model that bootstraps historical season trajectories under a negative-binomial observation model. Trajectories come from a local reimplementation of Richards-curve epidemic-scanner fits [28]: the Richards curve is a flexible, S-shaped growth-then-plateau curve long used to summarize an epidemic's rise and fall, fit separately to each historical season, from which we resample plausible future trajectories. Because it draws on the full spread of past seasons rather than assuming a fixed noise scale, this produces wide, well-calibrated intervals under distribution shift.

Ensembles. Models are combined by per-quantile median (Vincentization) and, separately, by inverse-WIS weights: each model's weight is proportional to $1/\overline{\text{WIS}}$ on the designated tuning season, so a model that scored better on that season counts for more, tuned only on the designated tuning season, so weights are never fit on the seasons on which the ensemble is reported.

4.6. Conformal recalibration

Section 2.3 gives a worked numeric example of this procedure's effect on one interval; here we give the full definition. For each central interval level L with bounds $[\ell_L, u_L]$ and median m, we compute a multiplicative widening factor s_L that makes the interval reach its nominal coverage on the tuning season. For each calibration unit, the multiple needed to just cover the observation is $r = (y - m)/(u_L - m)$ when $y > m$ and $r = (m - y)/(m - \ell_L)$ when $y < m$; s_L is the $L/100$ empirical quantile of r, a scale-free, multiplicative analogue of conformalized quantile regression that is robust to the large cross-state differences in scale. The interval is then widened about the median by s_L, and monotonicity across quantiles is restored by a rearrangement step. Factors are estimated on the tuning season and applied to all seasons; recalibrating the ensemble output outperforms recalibrating any single member. The same frozen, fold-1-derived factors are reused unchanged for the forecast phase, consistent with fitting recalibration weights only once and never on the season being forecast.

4.7. Size-aware recalibration

Let z_u be state u's log mean weekly case count from 2015 to the tuning season's training cutoff, standardized across the 26 states. Every quantile q of the state's forecast for a week with median m is mapped to $\exp\{\log(1+m)+s_u[\log(1+q)-\log(1+m)]\}-1$, with $s_u = \min\{1.4, \max\{0.4,\ a+b\,z_u\}\}$. The median is unchanged, $s_u < 1$ sharpens and $s_u > 1$ widens, and working on the $\log(1+\cdot)$ scale keeps lower bounds non-negative and stretches the skewed upper tail in proportion. (a, b) is chosen on a grid (a from 0.50 to 1.50, b from -0.20 to 0.40, steps of 0.05) to minimize, on the tuning season, the mean over states of $\log \overline{\mathrm{WIS}}_u$, subject to pooled mean WIS no greater than that of the unscaled forecast; $(1, 0)$, the identity, is always feasible. A geometric mean of WIS ratios against any fixed reference has the same minimizer, because the reference contributes a constant, so the criterion targets the organizers' R_v without using their model. The parameters and z_u are then frozen and applied unchanged to every season and to the forecast phase.

4.8. Reproducibility

All code, the backtesting harness, per-state scored predictions, and validated submission files for four official tracks (dengue and chikungunya, state and city level) are released. The pipeline is deterministic under a fixed seed, and an automated test suite covers leakage safety, scoring correctness, model persistence, and every model family. A provenance manifest records the exact data and code state behind each reported number. The Supporting Information provides the full feature list and frozen hyperparameters, the deep-learning architecture, the mechanistic derivation, per-state and per-fold tables, the ensemble-member and weight robustness analysis, and an item-by-item EPIFORGE 2020 reporting checklist [22].

Data and code availability

The pipeline was developed by team Neural Earth for the 3rd Infodengue–Mosqlimate Dengue Challenge. All code, data-processing pipeline, backtests, figures, and validated submissions are openly available at `https://github.com/kamrul28890/3rd_imdc_purdue_neuralearth`. The surveillance and covariate data are distributed by the challenge through the Mosqlimate platform.

Author contributions

Conceptualization, methodology, software, formal analysis, investigation, visualization, and writing of the original draft: M.K.K. Validation, data curation, and writing (review and editing): E.J.E. and A.A.H. All authors read and approved the final manuscript.

Competing interests

The authors have declared that no competing interests exist.

Funding

The authors received no specific funding for this work.

Ethics statement

This study used aggregate, publicly available disease-surveillance and covariate data distributed through the Infodengue and Mosqlimate platforms. No individual-level data were used and no ethical approval was required.

AI usage

Development of the analysis pipeline, statistical methods, code, and portions of this manuscript was assisted by Claude (Anthropic), an AI assistant, under the direction and review of the human authors. All methodological decisions, interpretations, and conclusions are the responsibility of the human authors, who reviewed and verified the code, analyses, and results.

Acknowledgments

We thank the organizers of the Infodengue–Mosqlimate Dengue Challenge and the Infodengue and Mosqlimate teams for curating and distributing the data.

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